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Student Submissions — CO2_AT...



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Student 10 **Dinith S R** 192522028RD

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Q1

C

#1

```
int main() {  
    int production, total = 0;  
    float average, efficiency;
```





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Q1. Electric Battery Manufacturing

Score: 0.00 TC: 0/3 passed

CODE

```
#include <stdio.h>

int main() {
    int production, total = 0;
    float average, efficiency;

    for (int month = 1; month ≤ 12; month++) {
        scanf("%d", &production);

        // Invalid production value
        if (production < 0) {
            continue;
        }

        // Equipment failure
        if (production == -999) {
            break;
        }

        total += production;

        // Incentive and Inspection Status
        if (production > 5000) {
            printf("Month %d: Incentive\n", month);
        } else if (production < 4000) {
            printf("Month %d: Inspection\n", month);
        } else {
            printf("Month %d: Target Met\n", month);
        }
    }

    average = total / 12.0;
    efficiency = (average / 5000.0) * 100;

    printf("Annual Production: %d\n", total);
    printf("Average Production: %.2f\n", average);
    printf("Production Efficiency: %.2f%%\n", efficiency);

    return 0;
}
```

INPUT

```
5000
5000
5000
5000
5000
5000
5000
5000
5000
5000
5000
5000
```

OUTPUT

```
Month 1: Target Met
Month 2: Target Met
Month 3: Target Met
Month 4: Target Met
Month 5: Target Met
Month 6: Target Met
Month 7: Target Met
Month 8: Target Met
Month 9: Target Met
Month 10: Target Met
Month 11: Target Met
Month 12: Target Met
Annual Production: 60000
Average Production: 5000.00
Production Efficiency: 100.00%
```

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Q2. Maintenance

Score: 0.00 TC: 0/3 passed

CODE

```
int i;
for(i = 1; i ≤ 15; i++)
{
    scanf("%d %d", &load, &maintenance);
}
```





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Q2. Maintenance

CODE

```
#include <stdio.h>

int main()
{
    int load , maintance;
    float percentage,available;
    int i;
    for(i = 1; i ≤ 15; i++)
    {
        scanf("%d %d", &load, &maintance);
        if(load > 500)
        {
            printf("Grid Failure\n");
            break;
        }
        if(maintance==1)
            continue;
        percentage = (load * 100.0)/ 500;
        available = 500 - load;
        printf("Load percentage = %.2f%%\n", percentage);
        printf("Available Capacity = %.2f MW\n", available);
        if(percentage ≤ 60)
            printf("Normal\n");
        else if(percentage ≤ 85)
            printf("High Load\n");
        else
            printf("Critical\n");
    }
    return 0;
}
```

INPUT	
250	0
350	1
470	0

[illegible]

Available Capacity = 30.00 MW



```
#include <stdio.h>
int main()
{
    int stock = 250000, doses, i;
```

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Load percentage = 94.00%

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Available Capacity = 30.00 MW
Critical

Q3. Medical

Score: 0.00 TC: 0/3 passed

CODE

```
#include <stdio.h>
int main()
{
    int stock = 250000, doses, i;
    char status[10];
    for(i = 1; i <= 25; i++)
    {
        printf("Hospital %d\n", i);
        printf("Enter Hospital Status (OPEN/CLOSED):");
        scanf("%s", status);
        if(status[0]=='C' || status[0]=='c')
        {
            printf("Hospital Closed\n");
            continue;
        }
        printf("Enter Doses Distributed: ");
        scanf("%d", &doses);
        stock = stock - doses;
        if(doses <= 10000)
            printf("DistributionStatus: Sufficient\n");
        else if(doses >= 80000)
            printf("Distribution Status: Minimum Requirement\n");
        else
            printf("Distribution Status: Emergency Supply Required\n");
        printf("Remaining Vaccine Stock = %d\n\n", stock);
        if(stock <= 0)
        {
            printf("Vaccine stock exhausted.\n");
            break;
        }
    }
    return 0;
}
```

INPUT

OPEN
12000
OPEN
8500
OPEN
7500

OUTPUT

Security Error: Disallowed code pattern detected.

Q4. Storage

— Not Submitted —

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Student: **ABHIR KUMAR** 192312459RD

Q1. Electric Battery Manufacturing

Score: 25.00 TC: 3/3 passed